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Field Case Study Report MAKSA App Validation on Fresh Equine Semen (Newcastle, Australia, 2025)

Date: December 2025
Location: Newcastle, Australia

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MSc in Animal Reproductive Physiology and PhD candidate in Biological Sciences, with over 12 years of applied experience in animal germ cell biology, cryopreservation, and *in vitro* embryo production. Experienced in translating laboratory-based reproductive technologies into practical outcomes, with a strong focus on data quality, method validation, and workflow optimisation.	Maksim Mershiev, The CEO&Co-Founder of Avare Biotech.
Signature:	Signature: Signed by: Maksim Mershiev 325996BF0B2B43A...

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Executive Summary / Conclusions

- MAKSA Technology demonstrated user-friendly operation performance for both fresh and frozen–thawed equine semen samples.
- Progressive motility values were consistently higher with MAKSA, likely reflecting the system's higher trajectory sensitivity settings, particularly in thawed samples.
- Total motility values reported by MAKSA were robust, though occasionally exceeding 100% in fresh samples, indicating potential for further calibration.
- Sperm concentration measured by MAKSA was higher than that reported by CASA, highlighting the benefit of software calibration using fresh-diluted equine semen datasets.
- Video acquisition followed AVARE guidelines with appropriate dilution factors, ensuring reproducibility and data transparency.

Objective

To validate the smartphone-based MAKSA (Mobile-Assisted Key Semen Analysis) technology powering Avare Biotech App for **fresh and frozen equine semen** evaluation against conventional CASA system.

Materials and Methods

Fresh semen was collected from mature, healthy Shetland and miniature crossbred pony stallions, immediately diluted, and processed in the laboratory using single-layer colloid centrifugation. Semen was collected with a pony-sized Missouri artificial vagina (Minitube, Ballarat, VIC, Australia) equipped with an in-line filter. Immediately after collection, ejaculates were diluted 2:1 (extender: semen) using EquiPlus extender (Minitube, LOT 30030104702), with all equipment kept at approximately 37 °C. Samples were transported to the laboratory in insulated containers at room temperature (20–25 °C) immediately. In the laboratory, six mL of extended semen was layered over three mL of EquiPure colloid (Tek-Event Pty Ltd., Rouse Hill, NSW, Australia) and centrifuged at 400 × g for 20 min. After removing the supernatant, the sperm pellet was resuspended in 1 mL of BWW and evaluated under CASA and normal microscope for MAKSA software. Frozen–thawed samples were thawed in a 37 °C water bath, centrifuged to remove excess extender and cryoprotectant, and then resuspended in 1 ml of BWW. The samples were subsequently analyzed using both CASA and MAKSA systems.

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Table 1MAKSA results comparison

Tests done	Key Semen Parameters	MAKSA	CASA	Difference, %
Stallion 1 A, Fresh Sample	Concentration M/ml	140.8	50.6	64.1
	Total Motility %	.	69.9	.
	Progressive Motility %	55.0	15.8	71.3
	Non-Progressive Motility %	86.0	.	.
Stallion 1 B, Fresh Sample	Concentration M/ml	172.8	63.6	63.2
	Total Motility %	.	87.6	.
	Progressive Motility %	78.0	33.4	57.2
	Non-Progressive Motility %	41.0	.	.
Stallion 2, Fresh Sample	Concentration M/ml	195.2	62.5	68.0
	Total Motility %	.	89.7	.
	Progressive Motility %	91.0	33.4	63.3
	Non-Progressive Motility %	69.0	.	.
Stallion 3, Fresh Sample	Concentration M/ml	166.4	75.5	54.6
	Total Motility %	.	72.8	.
	Progressive Motility %	64.0	29.9	53.3
	Non-Progressive Motility %	87.0	.	.
Stallion 1, Thawed Sample	Concentration M/ml	195.2	71.0	63.6
	Total Motility %	.	30.5	.
	Progressive Motility %	67.0	14.0	79.1
	Non-Progressive Motility %	12.0	.	.
Stallion 2, Thawed Sample	Concentration M/ml	70.4	24.0	65.9
	Total Motility %	.	30.0	.
	Progressive Motility %	55.0	11.0	80.0
	Non-Progressive Motility %	14.0	.	.
Stallion 3, Thawed Sample	Concentration M/ml	137.5	66.0	52.0
	Total Motility %	.	26.4	.
	Progressive Motility %	40.0	8.5	78.8
	Non-Progressive Motility %	9.0	.	.

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Discussion / Expert Insights

- Progressive motility demonstrated method-dependent variation, with CASA values lower than those generated by MAKSA (53.3–80.0%). In fresh samples, differences ranged from 53–71%, increasing to 79–80% in frozen–thawed samples. This highlights progressive motility as a high-value performance indicator that is particularly responsive to analytical settings and sample condition.
- Sperm concentration outputs from CASA were lower than MAKSA across all samples, with differences of 52.0–68.0%. These findings suggest an opportunity to further optimise MAKSA's concentration algorithms through targeted software training using fresh-diluted equine semen datasets, supporting improved alignment across platforms.
- Total motility values produced by MAKSA in fresh samples mostly exceeded 100%, indicating scope for parameter boundary refinement to enhance robustness and reporting confidence in high-motility scenarios.
- Overall, MAKSA delivered low-cost and user-friendly experience, producing consistent results across both fresh and frozen–thawed semen samples. Progressive motility emerged as a key differentiating metric, particularly in post-thaw evaluations where sensitivity and repeatability are critical.
- To further strengthen product performance and user adoption, recommended next steps include species-specific calibration presets, automated quality-control flags for out-of-range values, and expanded user guidance on optimal dilution and capture conditions.

Recommendations

Appendix

• Screenshot of the CASA results

• MAKSA PDF report samples

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Appendix – Equine semen video screenshots and the pdf report example of each sample tested

Stallion 1 A, Fresh Sample (CASA)

Animal ID: Ziggy Pilot 17.12.2025
Genetic Line:
Analysis Date & Time: 12/17/2025 13:8:31
Print Date: 12/17/2025
Collection Tech:
Lab Tech: HT1

Hamilton Thorne
100 Cummings Center, Suite 465E
Beverly, MA 01852

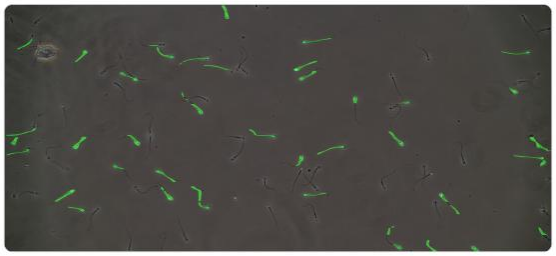
Equine Report

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	386	51	50.56	100
Static	116	15	15.19	30.10
Progressive	61	8	7.99	15.80
Motile	270	35	35.36	69.90
Slow	91	12	11.92	23.60

Stallion 1 A, Fresh Sample (MAKSA)

Analysis 17.12.2025
Animal: — **Name:** Ziggy **Age:** 12 **Breed:** Shetland Pony **Status:** Ready **Created:** 2025-12-17 02:35



Total objects:	88
Concentration (millions/ml^3):	140.80
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.002646
Progressive Motility:	86%
Non-Progressive Motility:	55%
Immotility:	1%

Stallion 1 B, Fresh Sample (CASA)

Animal ID: Ziggy Pilot 10.12.2025
Genetic Line:
Analysis Date & Time: 12/17/2025 13:19:7
Print Date: 12/17/2025
Collection Tech:
Lab Tech: HT1

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Beverly, MA 01852

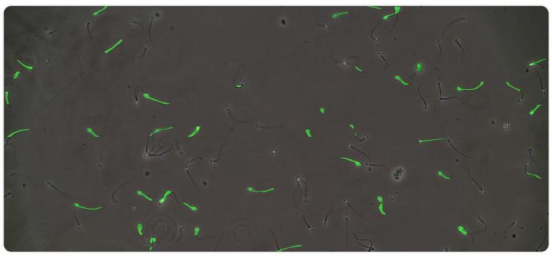
Equine Report

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	347	64	63.63	100
Static	43	8	7.88	12.40
Progressive	116	21	21.27	33.40
Motile	304	56	55.74	87.60
Slow	55	10	10.09	15.90

Stallion 1 B, Fresh Sample (MAKSA)

Analysis 17.12.2025
Animal: — **Name:** Ziggy 2 **Age:** 12 **Breed:** Shetland Pony **Status:** Ready **Created:** 2025-12-17 02:44



Total objects:	108
Concentration (millions/ml^3):	172.80
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.002358
Progressive Motility:	78%
Non-Progressive Motility:	41%
Immotility:	1%

Stallion 2, Fresh Sample (CASA)

Stallion 2, Fresh Sample (MAKSA)

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Animal ID: Bart Pilot 17.12.2025
Genetic Line:
Analysis Date & Time: 12/17/2025 13:15:13
Print Date: 12/17/2025
Collection Tech:
Lab Tech: HT1

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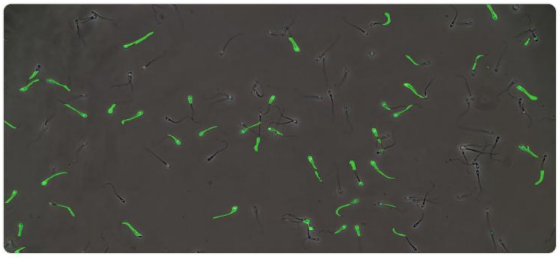
Equine Report

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	341	63	62.53	100
Static	35	6	6.42	10.30
Progressive	114	21	20.90	33.40
Motile	306	56	56.11	89.70
Slow	52	10	9.54	15.20

Stallion 3, Fresh Sample (CASA)

Analysis 17.12.2025
Animal: — **Name:** Bart **Age:** 8 **Breed:** Shetland Pony **Status:** Ready **Created:** 2025-12-17 02:45



Total objects:	122
Concentration (millions/ml^3):	195.20
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.001709
Progressive Motility:	91%
Non-Progressive Motility:	69%
Immotility:	1%

Stallion 3, Fresh Sample (MAKSA)

Animal ID: Pilot 17.12.2025
Genetic Line:
Analysis Date & Time: 12/17/2025 13:20:27
Print Date: 12/17/2025
Collection Tech:
Lab Tech: HT1

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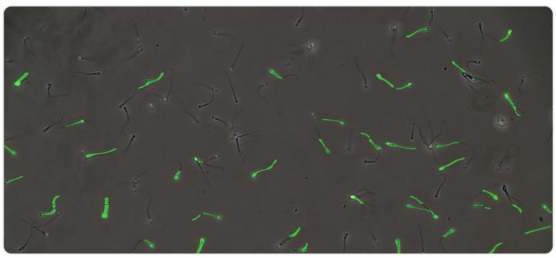
Equine Report

Motility

	Count	Sample M	Concentration M/ml	Percent Of Total
Total	412	76	75.55	100
Static	112	21	20.54	27.20
Progressive	123	23	22.55	29.90
Motile	300	55	55.01	72.80
Slow	62	11	11.37	15

Analysis 17.12.2025
Animal: — **Name:** Bart 2, 10th Dec **Age:** 8 **Breed:** Sgetland Pony **Status:** Ready **Created:** 2025-12-17 02:49



Total objects:	104
Concentration (millions/ml^3):	166.40
Dilution rate:	[1:1]
Volume of field of view (mm^3):	0.002408
Progressive Motility:	87%
Non-Progressive Motility:	64%
Immotility:	1%

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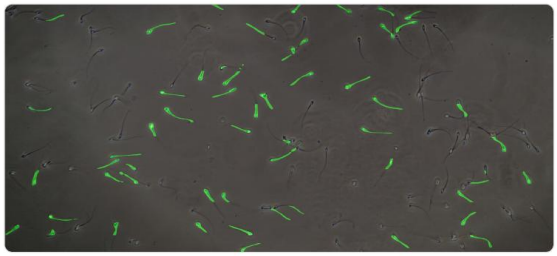
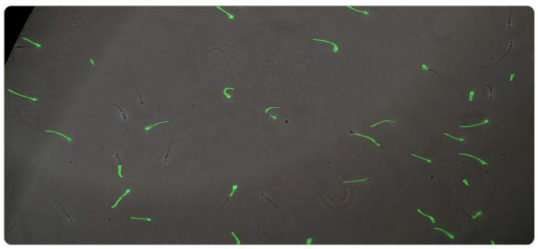
Stallion 1, Thawed Sample (CASA)	Stallion 1, Thawed Sample (MAKSA)
	Analysis 10.12.2025 **Animal:** — **Name:** Pat **Age:** 5 **Breed:** Shetland **Status:** Ready **Created:** 2025-12-10 02:07  Total objects: 122 Concentration (millions/ml^3): 195.20 Dilution rate: [1:1] Volume of field of view (mm^3): 0.002062 Progressive Motility: 67% Non-Progressive Motility: 12% Immotility: 21%
Stallion 2, Thawed Sample (CASA)	Stallion 2, Thawed Sample (MAKSA)
	Analysis 10.12.2025 **Animal:** — **Name:** Bart **Age:** 8 **Breed:** Shetland Miniature Crossbred Pony **Status:** Ready **Created:** 2025-12-10 02:03  Total objects: 44 Concentration (millions/ml^3): 70.40 Dilution rate: [1:1] Volume of field of view (mm^3): 0.002028 Progressive Motility: 55% Non-Progressive Motility: 14% Immotility: 31%
Stallion 3, Thawed Sample (CASA)	Stallion 3, Thawed Sample (MAKSA)

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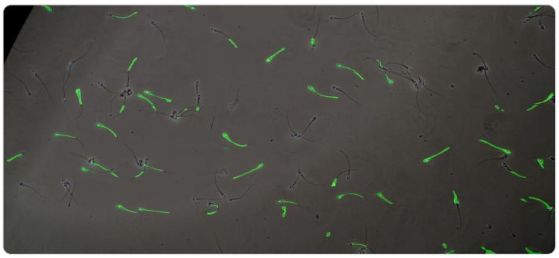
Analysis 10.12.2025

Animal: — **Name:** Nova **Age:** 6 **Breed:** Shetland Crosbred **Status:** Ready **Created:** 2025-12-10 02:06



Total objects:	**86**
Concentration (millions/ml^3):	**137.60**
Dilution rate:	**[1:1]**
Volume of field of view (mm^3):	**0.003988**
Progressive Motility:	**40%**
Non-Progressive Motility:	**9%**
Immotility:	**51%**